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Paleomagnetic results from Cretaceous sediments in Honduras: Tectonic implications

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ABSTRACT

Cretaceous sedimentary rocks from Honduras were collected in stratigraphic sequence for a paleomagnetic study. During Aptian-Albian time, Honduras rotated clockwise by 130° to a position rotated 165° relative to its present orientation and moved northward from 10°N to 30°N. In Cenomanian time, the rotation changed to counterclockwise and Honduras moved to ~20°N, close to its present latitude of 15°N. Near the end of the Cretaceous, Honduras was at 12°N in a position rotated 76° clockwise relative to its present orientation. These motions imply that, in Cretaceous time, Honduras was not part of the Caribbean plate as it exists today and that it originated in the Pacific Ocean rather than the Gulf of Mexico.

INTRODUCTION

All hypotheses about the tectonic evolution of the Caribbean suffer from the lack of data from Central America, particularly from the area of the present Caribbean plate. We report here results of a paleomagnetic study of Cretaceous rocks from Honduras which lies south of the Motagua-Polochic fault zone of Guatemala, the present boundary between the North American and Caribbean plates (Molnar and Sykes, 1969). Because most other paleomagnetic data from the Caribbean are from Cretaceous rocks and the Cretaceous pole position for North America is well known, a comparison is feasible both for data from within the Caribbean plate as well as between the Caribbean and North American plates.

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In recent years a series of detailed field studies (Horne and others, 1974; R. C. Finch and D. W. Curran, in prep.) has refined the basic stratigraphic framework of Mills and others (1967) and provided a suitable basis for a paleomagnetic study. The metamorphic basement of Honduras is of Paleozoic age or older and is overlain by Mesozoic sedimentary rocks. The Tertiary is represented by a thick sequence of volcanic rocks (Mills and others, 1967). All rock units discussed in this report are of Cretaceous age. The Aptian-Albian Yojoa Group consists of a basal shaly limestone member, the Cantarranas Formation, and a thick massive limestone, the Atima Formation. This group is conformably overlain by the Valle de Angeles Group, a thick red-bed sequence containing two possibly equivalent limestone members of the Cenomanian age, the Jaitique limestone (R. Finch and D. W. Curran, in prep.) and the Esquias Formation (Horne and others, 1974). The Valle de Angeles Group is unconformably overlain by Tertiary volcanic rocks.

PALEOMAGNETIC RESULTS

Although each outcrop was sampled stratigraphically, no long continuous sequences could be drilled owing to poor exposures and extensive faulting. All limestone samples were subjected to alternating field demagnetization, and the red-bed samples were thermally cleaned. A superconducting magnetometer was used to measure the remanent magnetization (Goree and Fuller, 1976).

The youngest rocks in this study are from the upper part of the Valle de Angeles Group. Because this red-bed unit does not contain any fossils, its stratigraphic position has only been bracketed between the Cenomanian and early to middle Tertiary. However, a more definite age assignment can be obtained paleomagnetically. Most of the rock units sampled are of Aptian through Cenomanian age, that is, from a time interval of normal magnetic polarity (McElhinny and Burek, 1971; Larson and Pittman, 1972). Indeed, all samples are of one polarity with the exception of

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TABLE 1. STRATIGRAPHIC SECTION AND PALEOMAGNETIC RESULTS

			TREAT	N	Dec	Inc	K	α95	Lat	Long	δp	δm
?	Valle de Angeles Group	11	TD500	9	284.9	21.9	134.2	4.5	17.0N	191.0	2.5	4.7
		10	TD500	34	103.3	-23.7	24.3	5.1	15.7N	191.3	2.9	5.4
Cenomanian	Esquias Fm.	9	AF200	21	290.2	34.5	89.9	3.4	23.5N	197.2	2.2	3.9
	?	8	AF100	16	273.1	34.1	20.0	8.5	7.5N	199.4	5.5	9.7
	Jaitique lst	7	AF100	15	271.3	24.4	57.5	5.1	4.5N	193.9	2.9	5.4
Albian	Valle de Angeles Group	6	TD500	35	266.4	24.7	56.4	3.3	0.1S	195.2	1.9	3.5
		5	TD500	18	227.0	35.5	29.5	6.5	32.4S	217.4	4.3	7.5
		4	TD500	8	195.2	33.9	96.3	5.7	53.5S	247.2	3.7	6.5
	Yojoa Group / Atima Fm.	3	AF200	25	214.6	49.3	48.5	4.2	34.1S	235.6	3.7	5.6
		2	AF150	51	315.6	35.3	43.6	3.1	47.5N	194.9	2.0	3.5
Aptian	Cantarranas Fm.	1	AF100	14	325.6	18.7	46.1	5.9	56.0N	177.3	3.2	6.2

Note: Stratigraphic section after R. C. Finch and D. W. Curran (in prep.). TREAT = treatment (alternating field or thermal demagnetization at specified value); N = number of samples; Dec = declination and Inc = inclination of remanent magnetization; K = precission parameter; α95 = 95% circle of confidence; Long = east longitude of paleomagnetic pole position; δp and δm are the semi-angles of the ellipse of confidence.

samples from an outcrop in the upper Valle de Angeles Group exposing 25 m of section that are of opposite polarity. We tentatively identify this reversely magnetized horizon with the first reversed zone after the long normal interval. If this assignment is correct, then these samples should be between 82 and 84 m.y. old (Larson and Pittman, 1972), defining a Coniacian age within the upper Valle de Angeles Group.

The paleomagnetic data together with statistical parameters are summarized in Table 1. The pole positions, which were computed under the assumption of a geocentric dipole field, were used to calculate the position of Honduras during part of the Cretaceous (Fig. 1). The longitude, of course, is undetermined. The data indicate that Honduras underwent large rotations but only minor changes in latitude. The first recorded movement in this study is a northerly motion coupled with a clockwise rotation. By the beginning of the Cenomanian (about 100 m.y. ago), Honduras was in a position rotated 165° clockwise relative to its present position and close to its present latitude. The rotation then changed to counterclockwise. By the end of the Cretaceous, Honduras was in a position rotated 76° clockwise relative to the meridian.

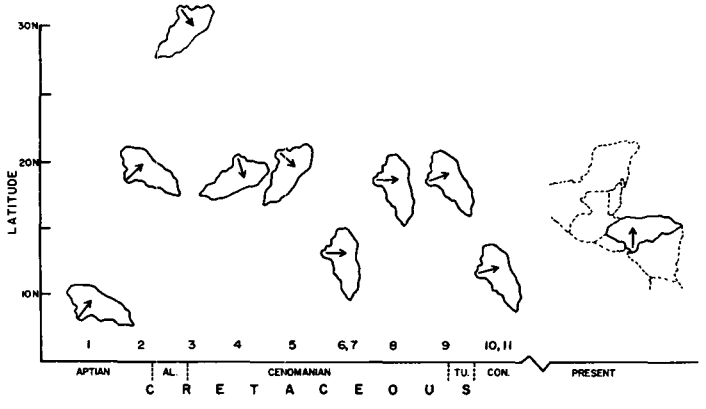


Figure 1. Schematic presentation of position of Honduras during part of Cretaceous Period relative to its present position.

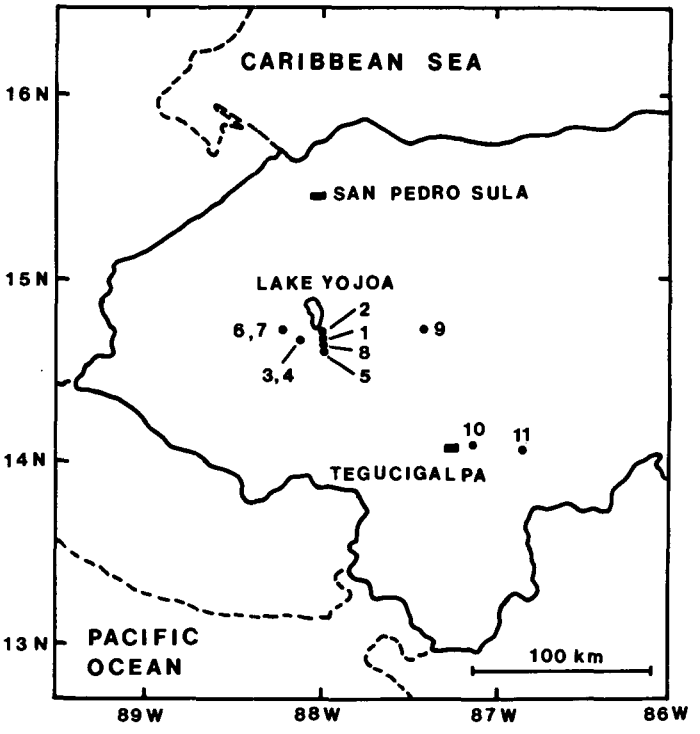


Figure 2. Collecting sites in Honduras. Numbers correspond to Table 1 and Figure 3.

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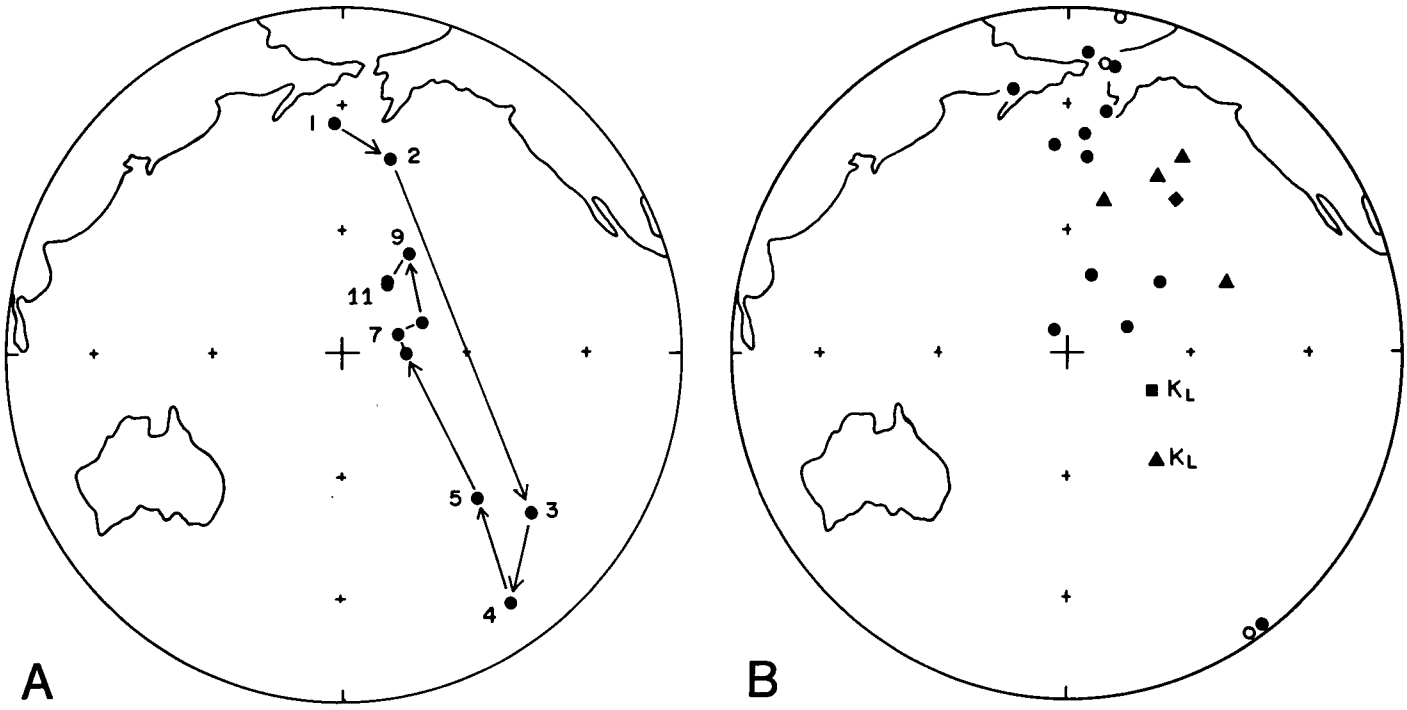


Figure 3. A. Cretaceous pole positions from Honduras. Arrows indicate progression of pole positions through time. See Table 1 for identification of numbers. B. Cretaceous pole positions from the Caribbean. ●○ = Jamaica (Watkins and Cambray, 1970); ▲ = Jamaica (Steinhauser and Vincenz, 1973); ■ = Colombia (MacDonald and Opdyke, 1972); ♦ = Puerto Rico (Fink and Harrison, 1972); open circles = opposite hemisphere. All poles are for Upper Cretaceous time except for two Lower Cretaceous poles (K_L).

The possibility needs to be considered that local tectonics such as block rotation might yield erroneous results. This seems highly unlikely because the samples were collected at many different, widely spaced outcrops (Fig. 2). For instance, the basal section of the Jaitique limestone (site 7) was sampled inside the Santa Barbara graben and all younger units to the east of it as far as 170 km away (sites 8 through 11). Yet all samples yield similar pole positions (see Fig. 3). It could be argued that the direction of magnetization at some sites, which we interpret to be of normal polarity, might actually be reversed. While only additional sampling, particularly from the Atima Formation, can test this alternative, the resulting plate motion would require rotations of similar magnitude but in a much more complex manner.

Throughout the time span covered by these samples, the North American plate was essentially stationary relative to the magnetic pole (McElhinny, 1973). Thus Honduras was clearly not part of the North American plate during this time. Neither was it part of the South American plate because that plate underwent only relatively small motions (McElhinney, 1973; Ladd, 1976). Recomputing the position of Honduras relative to the North American pole during Cretaceous time results in a 27° anticlockwise rotation of all positions shown in Figure 1.

DISCUSSION

The question arises whether the data from Honduras apply to the entire present Caribbean plate or, stated in a different way, whether the Caribbean plate existed as an entity in Cretaceous time. Paleomagnetic data from Puerto Rico (Fink and Harrison, 1972), Colombia (MacDonald and Opdyke, 1972), Jamaica (Vincenz and others, 1973), and Venezuela (Hargraves and Skerlec, 1976) are also interpreted in terms of large tectonic rotations. Whether these rotations are the same as seen in the Honduran data cannot be ascertained, because when dealing with fast motions it is necessary to have data with sufficient time resolution and time coverage, which is not the case for the published data. For example, the Jamaican data of Watkins and Cambray (1970) might very well indicate the same rotations as deduced from the Honduran results (Fig. 3A). But the age of the Jamaican rocks can only be specified as Upper Cretaceous and no relative age sequence is known. The other Cretaceous pole positions from the Caribbean (Fig. 3B) are in general agreement with the Honduran data; on closer inspection, however, some significant discrepancies exist. For instance, the Lower Cretaceous poles from Jamaica (Steinhauser and Vincenz, 1973) and Colombia (MacDonald and Opdyke, 1972) should be much closer to the pole positions derived from the Cantarranas and lower Atima Formations if they were to reflect the same plate motion.

We would like to speculate that the Caribbean plate did not exist as a single, rigid plate in Cretaceous time but rather that it consisted of several platelets. The Honduran platelet encompassed portions of southern Guatemala and northern Nicaragua, an area with pre-Mesozoic basement referred to as the Chortis block by Dengo and Bohnenberger (1969) and Dengo (1969) and possibly also the area of the Nicaraguan rise. We note that the basement in Central America south of northern Nicaragua is of Late Cretaceous age (Dengo, 1969) and that no land connection existed between South and North America until about Pliocene time (Bandy and Casey, 1973; Case and others, 1971). Nagle (1971) cited evidence that other parts of the Caribbean region also acted as individual tectonic units. A compilation of seismic epicenters for focal depths <50 km for the period 1970 through 1976 (T. Matumoto, personal commun.) shows that the present Caribbean plate boundaries are not as clearly defined as for many other plates, which might indicate that the welding of the platelets has not yet been completely accomplished. Perhaps one should look at the Carib-

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bean region as a buffer zone between the very large, surrounding plates. Similar suggestions have been made for other regions by Zijderveld and van der Voo (1973) and Beck (1976).

While clearly more data are needed before the tectonic evolution of the Caribbean can be understood, we present the following working model. At the beginning of the Cretaceous, Central America north of the Motagua-Polochic fault system was part of the North American plate, as evidenced by paleomagnetic data from the upper part of the Todos Santos Formation (uppermost Jurassic) in southern Mexico (Guerrero and Helsley, 1974) and northern Guatemala (our own preliminary results). The Gulf of Mexico was essentially formed and deposition of the Louann salt had ceased, implying open circulation. The northerly motion of the Chortis block during Early Cretaceous time favors a Pacific origin of the Chortis block as speculated by Malfait and Dinkelman (1972), Walper and Rowett (1972), and Moore and del Castillo (1974) rather than an origin in the Gulf of Mexico as suggested by Freeland and Dietz (1971). Our data imply substantial rotations of this area and are, therefore, incompatible with paleogeographic reconstructions based on simple strike-slip motion along the Motagua-Polochic fault (Hess and Maxwell, 1953; Pinet, 1972). Following the arguments of Maltait and Dinkelman (1972), we consider that the ancient Middle America Trench coincided with the present trench as far south as the Gulf of Tehuantepec and then bent easterly. The Motagua-Polochic fault zone might be the ancient trace of the trench. Geochemical studies of the El Tambor Formation (Lawrence, 1976) that crops out in the Motagua Valley show that the majority of the mafic rocks are normative tholeiites of mid-ocean ridge affinity although the rocks were metamorphosed probably in Late Cretaceous time. We speculate that the El Tambor Formation might be an uplifted part of oceanic crust similar to the Nicoya complex of Costa Rica (Henningsen and Weyl, 1967). During Late Cretaceous and early Tertiary time (MacDonald, 1976), the Chortis block rotated counterclockwise close to its present position. The Eocene seems to have been the time of major change in tectonic regime. Volcanism ceased in the Greater Antilles and began in the Lesser Antilles (Nagle, 1971); the mid-Cayman Ridge came into existence (Holcombe and others, 1973), which might signal the onset of motions as observed today (Matumoto and Latham, 1976; Plafker, 1976).

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